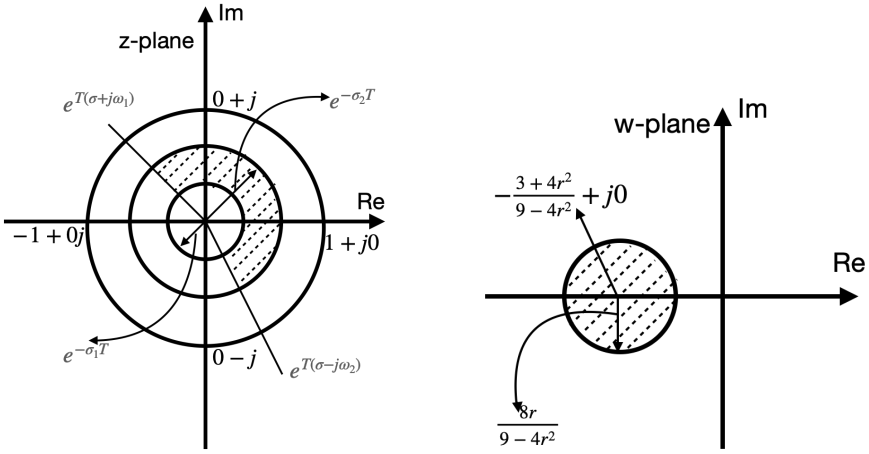


(a) z-plane (b) w-plane

Figure 1: z-plane and w-plane for question 1(A).

1(A). (i) Since $T = 2$, the primary strip is enclosed between $-\frac{\omega_s}{2} = -\frac{\pi}{T} < -1$ and $\frac{\omega_s}{2} = \frac{\pi}{T} > 1$. Thus, the shaded region in the s-plane lies in the primary strip. [2 marks]
The corresponding region in the z-plane is shown in Figure. 1(a). [3 marks]

(iii) Since $T = 2$, we have

$$z = \frac{1 + \frac{wT}{2}}{1 - 1 + \frac{wT}{2}} = \frac{1+w}{1-w}.$$

The shaded region in the z-plane is $|z - 1/2| \leq r$, which can be rewritten as

$$|\frac{1+w}{1-w} - 1/2| \leq r \Leftrightarrow |1 + 3w| \leq 2r|1 - w|$$

which is equivalent to

$$|w + \frac{3+4r^2}{9-4r^2}| \leq \frac{8r}{9-4r^2}.$$

Thus, the corresponding region at w-plane is a disk centered at $-\frac{3+4r^2}{9-4r^2} + j0$ with radius $\frac{8r}{9-4r^2}$, as shown in Figure. 1(b). [3 marks]

1(B). (i) According to the forward difference property of z-transform, we have

$$z^2X(z) - z^2x(0) - zx(1) + \alpha(zX(z) - zx(0)) + \beta X(z) = 0.$$

Thus,

$$X(z) = \frac{z^2 + \alpha z}{z^2 + \alpha z + \beta}x(0) + \frac{z}{z^2 + \alpha z + \beta}x(1).$$

[2 marks]

(ii) Let $\alpha = a + b$ and $\beta = ab$. Since $\alpha^2 - 4\beta > 0$, we have $a, b = -\frac{-\alpha \pm \sqrt{\alpha - 4\beta}}{2}$. Without loss of generality, let $a = -\frac{-\alpha + \sqrt{\alpha^2 - 4\beta}}{2}$ and $b = -\frac{-\alpha - \sqrt{\alpha^2 - 4\beta}}{2}$. Note that $a \neq b$.
Then,

$$X(z) = \frac{(z^2 + \alpha z)x(0) + zx(1)}{z^2 + (a+b)z + ab} = \frac{(z^2 + (a+b)z)x(0) + zx(1)}{(z+a)(z+b)}.$$

Expanding $X(z)/z$ into partial fractions, we have

$$\frac{X(z)}{z} = \frac{bx(0) + x(1)}{b-a}\frac{1}{z+a} + \frac{ax(0) + x(1)}{a-b}\frac{1}{z+b}.$$

Thus, we get

$$X(z) = \frac{bx(0)+x(1)}{b-a}\frac{1}{1+az^{-1}} + \frac{ax(0)+x(1)}{a-b}\frac{1}{1+bz^{-1}}.$$

The inverse z-transform of $X(z)$ gives

$$x(k) = \frac{bx(0)+x(1)}{b-a}(-a)^k + \frac{ax(0)+x(1)}{a-b}(-b)^k,\ k = 0, 1, 2, \ldots$$

where $a = -\frac{-\alpha+\sqrt{\alpha^2-4\beta}}{2}$ and $b = -\frac{-\alpha-\sqrt{\alpha^2-4\beta}}{2}$. [4 marks]

(iii) Assume that the initial values $x(0)$ and $x(1)$ are arbitrarily finite reals. The values $x(k)$ at $k = 2, 3, \ldots$ are bounded for any initial values $x(0)$ and $x(1)$ if and only if $|a| \leq 1$ and $|b| \leq 1$, that is

$$|-\alpha + \sqrt{\alpha^2-4\beta}| \leq 2 \text{ and } |\alpha + \sqrt{\alpha^2-4\beta}| \leq 2.$$

Recall that $\beta \leq \frac{a^2}{4}$. These conditions are equivalent to

$$\begin{cases} -2 \leq \alpha \leq 2 \\ \beta \leq \frac{a^2}{4} \\ \alpha + \beta \geq 1, \\ \alpha - \beta \geq 1. \end{cases}$$

The corresponds to the shaded region in Figure 2. [2 marks]

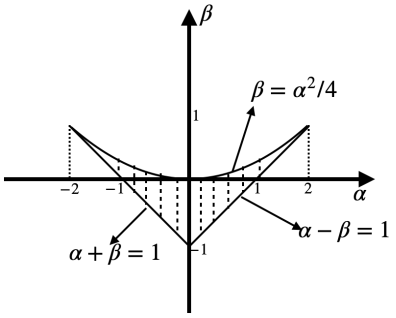


Figure 2: Region of α-β plane.

1(C). Consider the diagram in Figure 3.

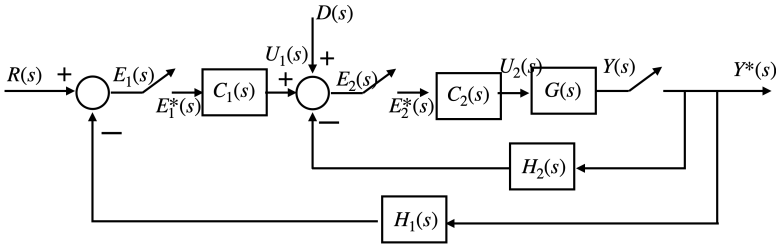


Figure 3: Diagram for question 1(C).

We have

$$\begin{aligned} E_1(s) &= R(s) - Y^*(s)H_1(s) \\ U_1(s) &= E_1^*(s)C_1(s) \\ E_2(s) &= U_1(s) + D(s) - Y^*(s)H_2(s) \\ Y(s) &= E_2^*(s)C_2(s)G(s). \end{aligned}$$

Applying the stared operator, we have

$$\begin{aligned}
E_1^*(s) &= R^*(s) - Y^*(s)H_1^*(s) \\
U_1^*(s) &= E_1^*(s)C_1^*(s) \\
E_2^*(s) &= U_1^*(s) + D^*(s) - Y^*(s)H_2^*(s) \\
Y^*(s) &= E_2^*(s)[C_2G]^*(s).
\end{aligned}$$

Then we have

$$\begin{aligned}
Y^*(s) &= ((R^*(s) - Y^*(s)H_1^*(s))C_1^*(s) + D^*(s) - Y^*(s)H_2^*(s))[C_2G]^*(s) \\
\Rightarrow Y(z) &= ((R(z) - Y(z)H_1(z))C_1(z) + D(z) - Y(z)H_2(z))[C_2G](z).
\end{aligned}$$

When $d(t) = 0$, we have

$$\frac{Y(z)}{R(z)} = \frac{C_1(z)[C_2G](z)}{1 + H_1(z)C_1(z)[C_2G](z) + H_2(z)[C_2G](z)}$$

[2 marks]

When $r(t) = 0$, we have

$$\frac{Y(z)}{D(z)} = \frac{[C_2G](z)}{1 + H_1(z)C_1(z)[C_2G](z) + H_2(z)[C_2G](z)}$$

[2 marks]

1(D). The characteristic equation to the closed-loop system under the unity feedback is

$$z^3 + 2z^2 + z + 2 = 0.$$

Applying the bilinear transformation $z = \frac{1+w}{1-w}$, we have

$$\frac{(1+w)^3}{(1-w)^3} + 2\frac{(1+w)^2}{(1-w)^2} + \frac{1+w}{1-w} + 2 = 0.$$

That is,

$$\frac{-2w^3 + 6w^2 - 2w + 6}{(1-w)^3} = 0.$$

Applying the Routh Theorem, we build the Routh table

n=3	-2	-2	0
n=2	6	6	0
n=1	4	0	0
n=0	6	0	0

Due to the sign change from -2 to 6, there exists at least one root whose real part is positive, which further implies there is a root outside the unit circle to the characteristic equation. Thus the closed-loop system is not stable. [5 marks]

2(A). Consider a unity feedback system with open-loop continuous-time transfer function

$$P(s) = \frac{K}{s(s+1)}$$

interconnected, by means of a ZOH and a sampler, to the digital controller $C(z) = 1$.

(i) The discrete-time equivalent transfer function is

$$\begin{aligned}
HP(z) &= Z\left(\frac{1 - e^{-sT}}{s}P(s)\right) = (1 - z^{-1})Z\left(\frac{P(s)}{s}\right) = K(1 - z^{-1})Z\left(\frac{1}{s^2(s+1)}\right) \\
&= K\frac{(T - 1 + e^{-T})z + 1 - e^{-T} - Te^{-T}}{(z - e^{-T})(z - 1)}.
\end{aligned}$$

[4 marks]

(ii) The characteristic equation is

$$z^2 + \left(K(T - 1 + e^{-T}) - e^{-T} - 1\right) z + e^{-T} + K(1 - e^{-T} - Te^{-T}) = 0.$$

Recall that the equation

$$z^2 + \alpha z + \beta = 0$$

has all roots in the unit circle if

$$\begin{cases} \alpha + \beta + 1 > 0 \\ 1 - \beta > 0 \\ 1 - \alpha + \beta > 0 \end{cases}$$

Applying these condition yields that K should fulfill

$$\begin{cases} K(T - 1 + e^{-T}) - e^{-T} - 1 + K(1 - e^{-T} - Te^{-T}) + e^{-T} > 0 \\ 1 - K(1 - e^{-T} - Te^{-T}) - e^{-T} > 0 \\ 1 - K(T - 1 + e^{-T}) + e^{-T} + 1 + K(1 - e^{-T} - Te^{-T}) + e^{-T} > 0 \end{cases}$$

2(B). (i) The open-loop transfer function is

$$C(s)P(s) = \frac{K}{(s + a)(s + b)}$$

.

The transfer function for the closed-loop continuous-time system is

$$G(s) = \frac{C(s)P(s)}{1 + C(s)P(s)} = \frac{K}{s^2 + (a + b)s + K + ab}.$$

The asymptotic stability of this system depends on the roots of the characteristic equation $s^2 + (a + b)s + K + ab = 0$. Applying the Routh test, its roots lie in the left half of the s-plane if and only if $a + b > 0$ and $K + ab > 0$. [3 marks]

(ii) The equivalent discrete-time model of the plant is

$$\begin{aligned} HP(z) &= Z\left(\frac{1 - e^{-sT}}{s} P(s)\right) = (1 - z^{-1}) Z\left(\frac{P(s)}{s}\right) = (1 - z^{-1}) Z\left(\frac{1}{s(s + a)}\right) \\ &= (1 - z^{-1}) \frac{1}{a} \left(\frac{1}{1 - z^{-1}} - \frac{1}{1 - e^{-aT} z^{-1}}\right) = \frac{1}{a} \frac{1 - e^{-aT}}{z - e^{-aT}}. \end{aligned}$$

[4 marks]

(iii) To determine $C(z)$ with the impulse response invariance method, we compute the z-transform of $C(s)$, that is,

$$C_I(z) = Z(C(s)) = \frac{Kz}{z - e^{-bT}}.$$

[2 marks]

(iv) The root locus of $C_I(z)HP(z)$ is shown in Figure 4. The break-away point is $e^{-\frac{1}{2}(a+b)T}$ and the break-in point is $-e^{-\frac{1}{2}(a+b)T}$. [2 marks]

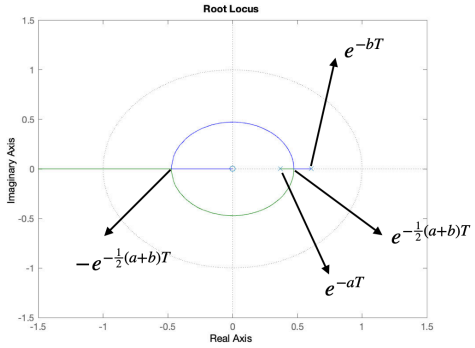


Figure 4: The root locus of $C_I(z)HP(z)$.

The characteristic equation when using $C_I(z)$ is

$$(z - e^{-aT})(z - e^{-bT}) + \frac{K}{a}(1 - e^{-aT})z = 0$$

which is equivalent to

$$z^2 + \left(\frac{K}{a}(1 - e^{-aT}) - e^{-aT} - e^{-bT}\right) z + e^{-aT-bT} = 0. \tag{1}$$

Recall that the equation

$$z^2 + \alpha z + \beta = 0$$

has all roots in the unit circle if

$$\begin{cases} \alpha + \beta + 1 > 0 \\ 1 - \beta > 0 \\ 1 - \alpha + \beta > 0 \end{cases}$$

Applying these condition to (1) yields

$$\frac{a(e^{-aT} + e^{-bT} - 1 - e^{-aT-bT})}{1 - e^{-aT}} < K < \frac{a(1 + e^{-aT-bT} + e^{-aT} + e^{-bT})}{1 - e^{-aT}}.$$

[3 marks]

(v) There are two different ranges of K for ensuring the asymptotically stability. In particular, the discretization introduces a maximum on K. For $C_I(z)$, the maximum and minimum gains depend on a and b marginally. Note that

$$\lim_{T\to 0} \frac{a(e^{-aT} + e^{-bT} - 1 - e^{-aT-bT})}{1 - e^{-aT}} = \lim_{T\to 0} \frac{a(-ae^{-aT} - e^{-bT} + (a+b)e^{-aT-bT})}{ae^{-aT}} = 0.$$

When $T \to 0$ the maximum gain approaches to $+\infty$ where the minimum gain approaches to 0, greater than $-ab$ as in the continuous-time case. [2 marks]

3 (i) For continuous-time linear system, the reachability is equivalent to controllability. The reachability matrix $\begin{bmatrix} G & FG \end{bmatrix} = \begin{bmatrix} 0 & a \\ 1 & 0 \end{bmatrix}$ and the observability matrix is $\begin{bmatrix} H \\ HF \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -a & 0 \end{bmatrix}$ Both matrices are full-rank since a is a non-zero constant. Thus, the continuous-time system is controllable and observable. [4 marks]

(ii) The matrix A is $A = e^{FT}$. It follows that

$$\begin{aligned} e^{Ft} &= \mathcal{L}^{-1}[(sI - F)^{-1}] = \mathcal{L}^{-1}\left[\begin{bmatrix} s & -a \\ a & s \end{bmatrix}^{-1}\right] \\ &= \mathcal{L}^{-1}\left[\begin{bmatrix} \frac{s}{s^2+a^2} & \frac{a}{s^2+a^2} \\ -\frac{a}{s^2+a^2} & \frac{s}{s^2+a^2} \end{bmatrix}\right] == \begin{bmatrix} cos(at) & sin(at) \\ -sin(at) & cos(at) \end{bmatrix} \end{aligned}$$

Thus, we have

$$A = e^{FT} = \begin{bmatrix} cos(aT) & sin(aT) \\ -sin(aT) & cos(aT) \end{bmatrix}.$$

[1 marks]

The matrix B is

$$\begin{aligned} B &= \int_0^T e^{Ft} dt G = \int_0^T \begin{bmatrix} cos(at) & sin(at) \\ -sin(at) & cos(at) \end{bmatrix} dt G \\ &= \begin{bmatrix} \frac{1-\cos(aT)}{a} \\ \frac{\sin(aT)}{a} \end{bmatrix} \end{aligned}$$

[2 marks]

The matrix C is the same as H, i.e., $C = \begin{bmatrix} 0 & 1 \end{bmatrix}$. [1 marks]

(iii) (1) The reachability matrix is

$$\begin{bmatrix} B & AB \end{bmatrix} = \begin{bmatrix} \frac{1-\cos(aT)}{a} & \frac{\cos(aT)-\cos^2(aT)+\sin^2(aT)}{a} \\ \frac{\sin(aT)}{a} & \frac{\sin(aT)(-1+2\cos(aT))}{a} \end{bmatrix} = \begin{bmatrix} \frac{1-\cos(aT)}{a} & \frac{\cos(aT)-\cos^2(aT)+\sin^2(aT)}{a} \\ \frac{\sin(aT)}{a} & \frac{\sin(aT)(-1+2\cos(aT))}{a} \end{bmatrix}.$$

This matrix is full-rank if and only if $\frac{\sin(aT)}{a} \neq 0$, which is equivalent to $aT \neq k\pi$, $k = \pm 1, \pm 2, \ldots$.

[2 marks]

(2) The observability matrix is $\begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\sin(aT) & \cos(aT) \end{bmatrix}$.

This matrix is full-rank if and only if $\sin(aT) \neq 0$, which is equivalent to $aT \neq k\pi$, $k = \pm 1, \pm 2, \ldots$.

[2 marks]

(3) The transfer function of the discrete-time linear system is

$$\frac{Yz}{U(z)} = C(zI - A)^{-1}B = \frac{1}{a} \frac{\sin(aT)(z-1)}{z^2 - 2\cos(aT)z + 1}.$$

[2 marks]

The transfer function is nonzero if and only if $\sin(aT) \neq 0$, which is equivalent to $aT \neq k\pi$, $k = \pm 1, \pm 2, \ldots$. [2 marks]

3(B). Consider the system

$$x(k+1) = Ax(k) + Bu(k)$$

and the discounted performance index

$$J_N = \gamma^N x^\top(N) S x(N) + \sum_{k=0}^{N-1} \gamma^k \left(x^\top(k) Q x(k) + u^\top(k) R u(k) \right)$$

with $0 < \gamma \leq 1$ is a constant, Q and S are positive definite or positive semidefinite matrices, and R is a positive definite matrix.

(i) Let the cost-to-go be $V_k(x)$:

$$V_k(x(k)) = \min_{u(k),\ldots,u(N-1)} \gamma^N x^\top(N) S x(N) + \sum_{i=k}^{N-1} \gamma^i \left(x^\top(i) Q x(i) + u^\top(i) R u(i) \right).$$

[3 marks]

(ii) When $k = N$, $V_k(x) = x^\top P_N x$ with $P_N = \gamma^N S$.

Assume that $V_{k+1}(x) = x^\top P_{k+1} x$. Let us derive the cost-to-go $V_k(x_k)$:

$$\begin{aligned}
V_k(x(k)) =] & \min_{u(k)} V_{k+1}(x(k+1)) + \gamma^k \left(x^\top(k) Q x(k) + u^\top(k) R u(k)\right)] \\
&= \min_{u(k)} [x^\top(k+1) P(k+1) x(k+1) + \gamma^k \left(x^\top(k) Q x(k) + u^\top(k) R u(k)\right)] \\
&= \min_{u(k)} [(Ax(k) + Bu(k))^\top P(k+1)(Ax(k) + Bu(k)) + \gamma^k \left(x^\top(k) Q x(k) + u^\top(k) R u(k)\right)] \\
&= \min_{u(k)} [x^\top(k)(A^\top P(k+1) A + \gamma^k Q) x(k) + x^\top(k) A^\top P(k+1) B u(k) \\
&\qquad + u^\top(k) B^\top P(k+1) A x(k) + u^\top(k)(\gamma^k R + B^\top P(k+1) B) u(k)]
\end{aligned}$$

[2 marks]

The optimal controller is $u(k) = -K(k)x(k)$ where

$$K(k) = (\gamma^k R + B^\top P(k+1) B)^{-1} B^\top P(k+1) A.$$

[2 marks]

And the matrix $P(k)$ is

$$P(k) = A^\top P(k+1) A + \gamma^k Q - A^\top P(k+1) B (\gamma^k R + B^\top P(k+1) B)^{-1} B^\top P(k+1) A.$$

with $P(N) = \gamma^N S$. [2 marks]